

Urban Wünsch

Assistant Professor

Researcher working on organic compounds dissolved in water — disentangling the composition of complex environmental samples to understand the fate of Dissolved Organic Matter (DOM) in aquatic systems. As of late 2026: 34 publications (10 as lead), H-index 19, 2100+ citations.

✉ urban.wunsch@chalmers.se 📍 Gothenburg, Sweden, SE

Education

Technical University of Denmark

PhD

2018

University of Applied Sciences Bremerhaven

M.Sc. Biotechnology

2014

University of Applied Sciences Bingen

B.Sc. Biotechnology

2011

Experience

Chalmers University of Technology, Architecture and Civil Engineering

Assistant Professor

2025 – Present

DTU Aqua, National Institute for Aquatic Resources

Assistant Professor

2024 – 2025

Postdoc

2020 – 2024

Chalmers University of Technology, Architecture and Civil Engineering

Postdoc

2018 – 2020

Technical University of Denmark, National Institute for Aquatic Resources

PhD student

2014 – 2017

Teaching

Blue Bioeconomy Innovation

2023 – 2024

Course responsible

M.Eng, ~40 students

Ocean science & technology

2015 – 2025

Teaching topics in master-level courses

Machine learning for spectroscopic data analysis

2018 – Present

Co-organizer of online course

For Master- and PhD-students

Introduction to Chemistry & Urban spaces and functions for civil engineers

2018 –
2020

Teaching, bachelor-level courses

Large classrooms

Supervision

Julia Iwan (PhD)

2025 – Present

Main supervisor

Frederik Lenk (B.Sc.)

2024 – 2024

Main supervisor

Special course

Maria Papadimitraki (Research Assistant)

2023 – 2023

Co-supervisor

University of Southern Denmark

Emilie Tage Andresen (B.Sc.)

2022 – 2022

Co-supervisor

DTU Chemistry (Denmark)

Linea Gry Ebbesen (B.Eng.)

2022 – 2022

Co-supervisor

DTU Environment (Denmark)

Laila Vinther & Cecilie Hjelm Hvas Hansen (M.Sc.)

2021 – 2021

Co-supervisor, voluntary special course

DTU Environment (Denmark)

Evelyn Namuga Kasule (B.Sc.)

2021 – 2021

Co-supervisor

DTU Environment (Denmark)

Signe Melbye Andersen (M.Sc.)

2020 – 2022

Co-supervisor

DTU Aqua (Denmark)

Anders Dalhoff Bruhn Jensen (PhD)

2019 – 2023

Co-supervisor

DTU Aqua (Denmark)

Xianyu Kong (PhD)

2019 – 2022

PhD committee member

Alfred Wegener Institute (Germany)

Nashita Moona (PhD)

2018 – 2021

Co-supervisor

Chalmers (Sweden)

Skills

Dissolved organic matter characterization

Excitation-Emission Matrix (EEM) spectroscopy

Absorbance spectroscopy

High Performance Size Exclusion Chromatography
(HPSEC)

Fourier Transform Ion Cyclotron Resonance Mass
Spectrometry (FT-ICR-MS)

Chemometrics

Principal Component Analysis (PCA)

Parallel Factor Analysis (PARAFAC & PARAFAC2)

N-way Partial Least Squares (N-PLS)

Advanced Coupled Matrix and Tensor Factorization
(ACMTF)

Research grants

Dissolved Organic Matter Biogeochemistry - innovative methods and sensors

2025

Strategic funding, Chalmers thematic area Ocean

2025-2029, Principal Investigator

A new approach to laccase discovery from the sea	2023
Independent Research Fund Denmark	
2.5 million DKK, named applicant	
New eyes on old GAC	2022
FORMAS, Sweden	
146,000 SEK, co-applicant	
Oxidation Demand and Rate (ODaR) as a sensitive proxy of biostability in drinking water	2021
FORMAS, Sweden	
26,000 SEK, co-applicant	
Enhancing Biogeochemical Fingerprints of Natural Organic Matter with Data Fusion	2019
Åforsk Foundation	
400,000 SEK, main applicant	
An optical sensor for reducing biocide dependence in chemical manufacturing	2019
Vinnova, Swedish Governmental Agency for Innovation Systems	
474,000 SEK, co-applicant	
2-year postdoctoral scholarship	2019
Carl Tryggers Stiftelse	
552,000 SEK, main author	

Conferences

Photosensitivity of fluorescent dissolved organic matter across the arctic ocean	2023
Aquatic Sciences Meeting 2023, Mallorca, Spain	
Combining Physical and Mathematical Chromatography to Decipher the Molecular Fingerprints of Dissolved Organic Matter	2021
20th Conference of the International Humic Substances Society (online)	
Mathematical chromatography deciphers the molecular fingerprints of dissolved organic matter	2021
Aquatic Sciences Meeting 2021 (online)	

Membership in Scientific Communities

**Association for the Sciences of
Limnology and Oceanography (ASLO)**

Member since 2014

**International Humic Substances Society
(IHSS)**

Member since 2018